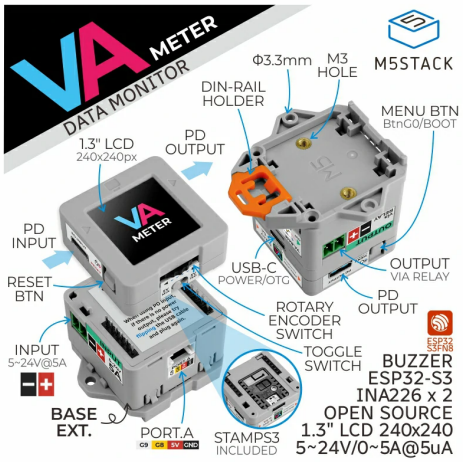
VAMeter

SKU:K136







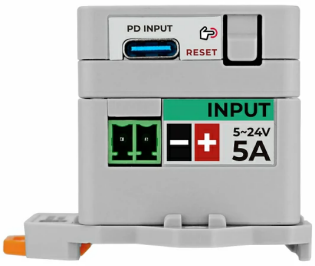












Description

VAMeter is a professional, programmable, high-precision power meter designed for accurate voltage and current measurements. Powered by the **Stamp-S3** module, it features Wi-Fi connectivity and two cascaded **INA226** current/voltage sensing circuits, delivering current detection resolutions of 2.5 µA and 250 µA. The measurement range supports 0-5 A and 5-24 V (1-24 V in isolation mode).

The device offers flexible power-supply options through the USB-C port or the Base terminal, with built-in power and signal isolation. A slide switch lets users choose between isolated measurements powered via USB and non-isolated measurements powered directly from the test source, meeting various application needs.

The included expansion BASE provides relay on/off control and a Grove port for secondary development. Additional features include OTA firmware upgrades and USB MSC/OTG support, allowing users to view and edit data and configuration files stored in the main controller' s file system from a PC.

M5Stack also supplies a free online data API for uploading test data to the **EZData** platform, enabling remote access and visualization and greatly simplifying IoT application development.

VAMeter is a highly practical and developer-friendly instrument, ideal for projects requiring precise current and voltage measurements, as well as for industrial automation, equipment monitoring, smart grid, and related fields.

Tutorial



Quick Start

This tutorial introduces usage examples of VAMeter.

Features

- High-precision current detection (2.5 µA / 250 µA resolution switching)
- Wide voltage input (5-24 V)
- Cloud data display via EZData
- PD input (no PD trigger built in)
- Multiple power-supply methods (PD INPUT / Stamp-S3 / external terminal block)
- Power and signal isolation
- USB MSC/OTG support
- Open-source hardware & software
- Wireless OTA upgrade
- Development Platform
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x VAMeter
- 1 x VAMeter Base
- 2 x 3.81-2P Terminal Block
- 4 x Hanger (Small)

Applications

- Industrial automation
- Equipment monitoring
- Power testing & analysis
- IoT devices
- Remote monitoring & data acquisition

Specifications

Specification	Parameter
SoC	ESP32-S3FN8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Power Detection Chip	INA226
Flash	8MB
Wi-Fi	2.4 GHz Wi-Fi
INA226 Communication Address	Dual detection chips INA226: 0x40 / 0x41
Display	1.3 inch @ 240 × 240 pixels
Display Driver	ST7789
PD Input Power	5-24 V @ 0-5 A
Current Resolution	2.5 µA / 250 µA
Current Sampling Err	Small current (1-100 mA): 1.01 % Large current (100 mA-5 A): 0.94 % µA-level current: 0.69 %
Voltage Error	Measurement error: 0.11 %
Power Consumption	Stamp-S3 USB powered: DC 5 V / 123.5 mA PD INPUT: DC 5 V / 118 mA DC 12 V / 66.8 mA DC 24 V / 26 mA
PD OUTPUT Capability	DC 24 V @ 3 A 68 ℃ DC 24 V @ 5 A 92 ℃
User Interface	Rotary encoder switch, Reset button, Boot button, onboard buzzer
Relay	DC 5 - 24 V @ 5 A
Operating Temp.	0 - 40 °C
Product Size	51.5 × 48.0 × 40.5 mm
Product Weight	43.4 g
Package Size	160.9 × 93.0 × 31.6 mm
Gross Weight	68.4 g

Learn

Download Mode

Download Mode

To enter download mode, press and hold the G0 button on VAMeter before power-on, then release it after powering up.

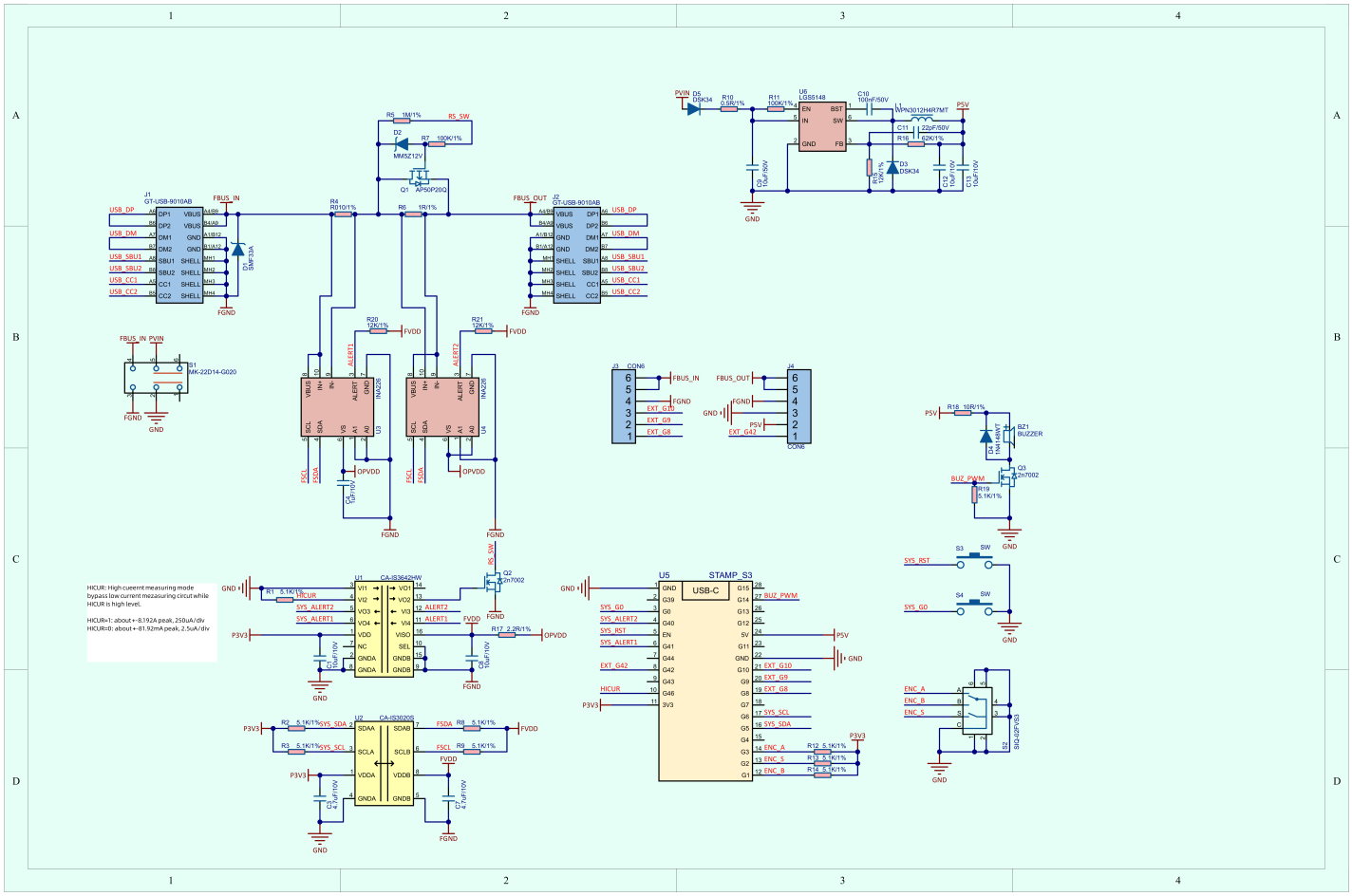


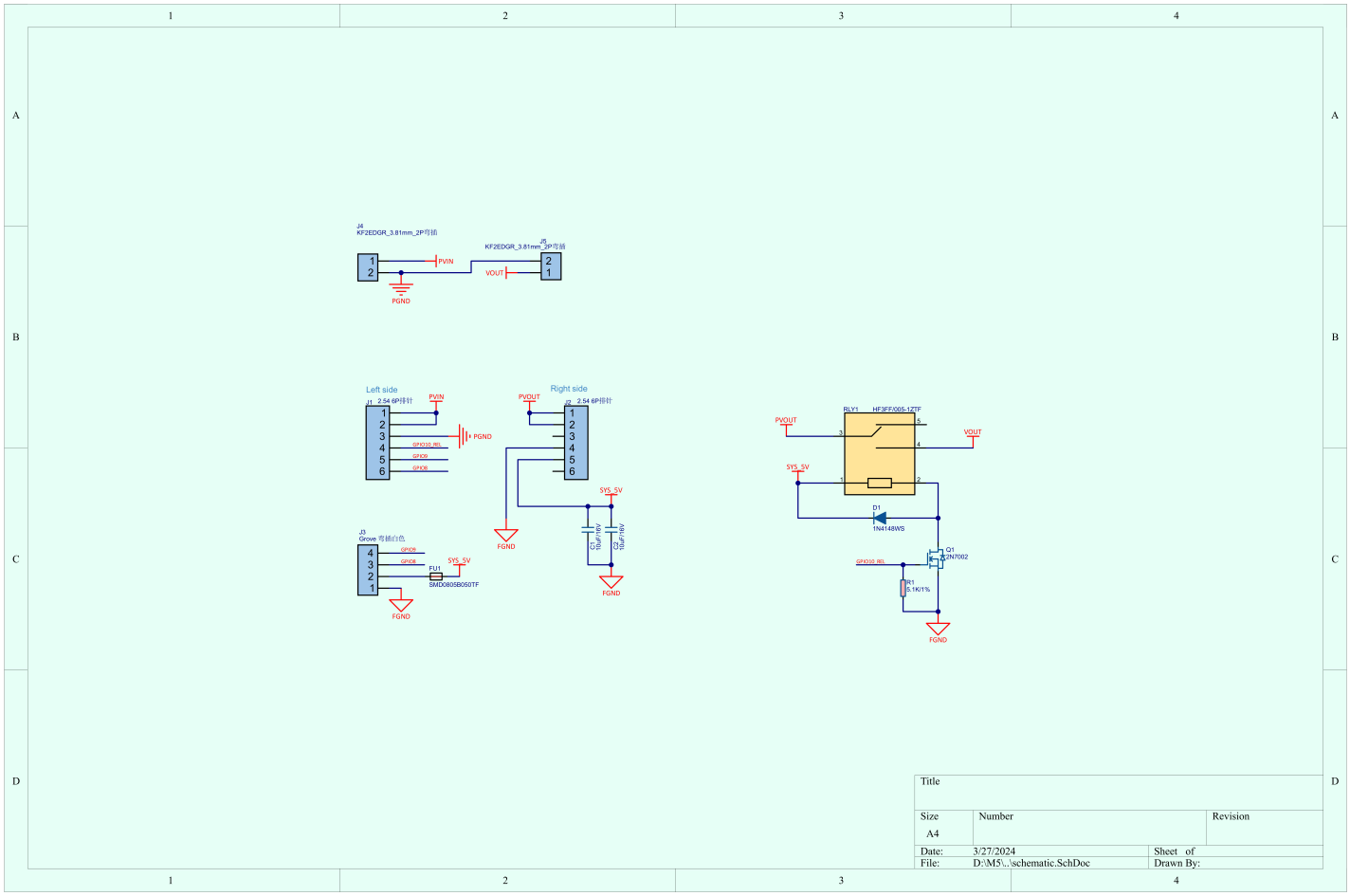
M5 EZData Cloud Platform Example



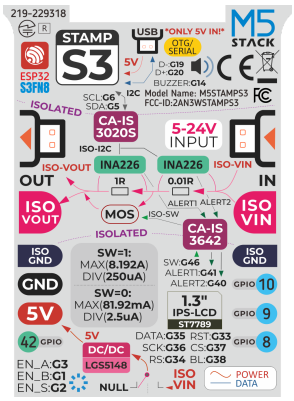
Schematics

- VAMeter Schematics PDF
- VAMeter Base Schematics PDF


HICUR: High cuerent measuring mode
bypass low current measuring circuit while
HICUR is high level.
HICUR=1: about +-8.192A peak, 250uA/div
HICUR=0: about +-81.92mA peak, 2.5uA/div
STAMP_S3
USB-C
CA-IS3642HW
CA-IS3020S
INA226
GT-USB-9010AB
GS5148
MMSZ12V
AP50P20Q
DSK34
MK-22D14-G020
ZN7002
8UZZER
SMD0805B050TF
SHD-02PVS3
KF2EDGR_3.81mm_2P弯插
2.54 6P排针
HF3FF/005-1ZTF
1N4148WS


KF2EDGR_3.81mm_2P弯插
Left side
Right side
2.54 6P排针
Grove 弯插白色
HF3FF/005-1ZTF
1N4148WS
Title
Size
A4
Number
Revision
Date: 3/27/2024
File: D:\M5\..\schematic.SchDoc
Sheet of
Drawn By:

VAMeter Internal Block Diagram



PinMap

VAMeter (INA226 & BUZZER)

> **Note**
>
> INA226 (0x40) and INA226 (0x41) are used for measurements at different resolutions and ranges.

ESP32S3FN8	G5	G6	G41	G40	G14
INA226 (0x40)	FSDA	FSCL	ALERT1		
INA226 (0x41)	FSDA	FSCL		ALERT2	
BUZZER					BUZ_PWM

VAMeter Expansion Pins

ESP32S3FN8	G10	G9	G8	G42
Expansion	EXT_G10	EXT_G9	EXT_G8	EXT_G42

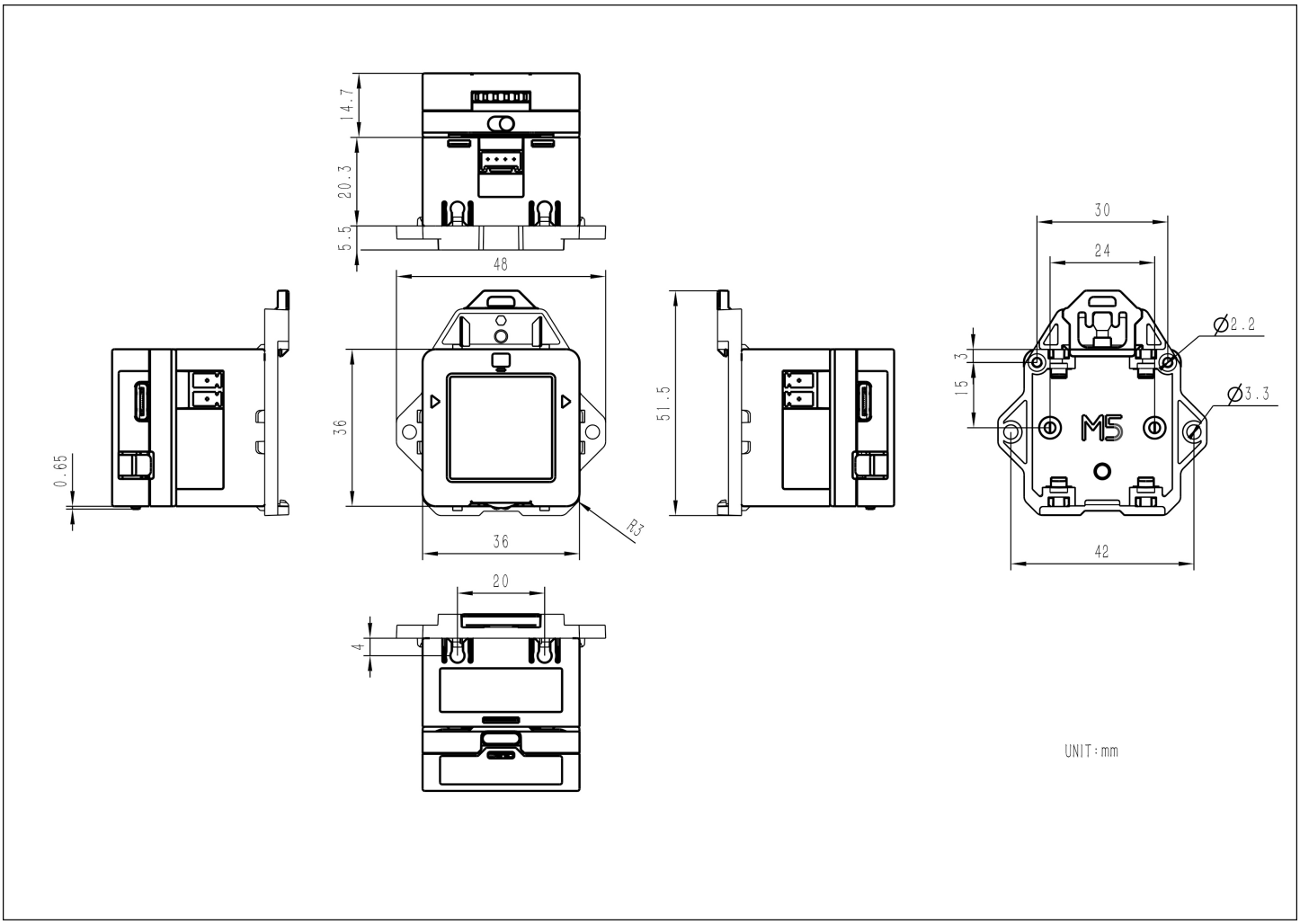
VAMeter Base

ESP32S3FN8	G10
Relay	G10_REL

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G8	G9

Model Size



Structure

- VAMeter Structure Files

PCB

- VAMeter Base Board PCB Size

Datasheets

- INA226

Softwares

Quick Start

- VAMeter Usage Example

ESP-IDF

- VAMeter Factory Firmware

Easyloader

Easyloader	Download	Note
VAMeter Firmware Easyloader	download	/

Other

- VAMeter Factory Firmware Restore Guide

Video

Product Comparison

PRODUCT	VAMETER	KWS-10VA	KWS-V20
Picture			
Input Voltage	DC 5-24V	DC 3-9V	DC 4-20V
Measure current	5A	3A	3A
Current resolution	2.5uA/250uA	0.01A	0.01A
Current Accuracy	Low Current（1-50mA）：1.01% Large Current（100mA-5A）：0.94% uA level current：0.69%	1%	1%
Voltage accuracy	0.11%	1%	1%